

Lab Assignment No.	1
Title	Set up and Configuration Hadoop Using CloudEra/ Google Cloud BigQuery. Databricks Lakehouse Platform. Snowflake. Amazon Redshift
Roll No.	
Class	BE
Date of Completion	
Subject	Computer Laboratory-IV : Big Data Analytics
Assessment Marks	
Assessor's Sign	

EXPERIMENT NO. 1 A (Group A)

Aim: Set up and Configuration Hadoop Using CloudEra/ Google Cloud BigQuery. Databricks Lakehouse Platform. Snowflake. Amazon Redshift

Theory:

Hadoop software can be installed in three modes of

Hadoop is a Java-based programming framework that supports the processing and storage of extremely large datasets on a cluster of inexpensive machines. It was the first major open source project in the big data playing field and is sponsored by the Apache Software Foundation.

Hadoop-2.7.3 is comprised of four main layers:

- **Hadoop Common** is the collection of utilities and libraries that support other Hadoop modules.
- **HDFS**, which stands for Hadoop Distributed File System, is responsible for persisting data to disk.
- **YARN**, short for Yet Another Resource Negotiator, is the "operating system" for HDFS.
- **MapReduce** is the original processing model for Hadoop clusters. It distributes work within the cluster or map, then organizes and reduces the results from the nodes into a response to a query. Many other processing models are available for the 2.x version of Hadoop.

Hadoop clusters are relatively complex to set up, so the project includes a stand-alone mode which is suitable for learning about Hadoop, performing simple operations, and debugging.

Procedure:

we'll install Hadoop in stand-alone mode and run one of the example MapReduce programs it includes to verify the installation.

Prerequisites:**Step1: Installing Java 8 version.**

Openjdk version "1.8.0_91"

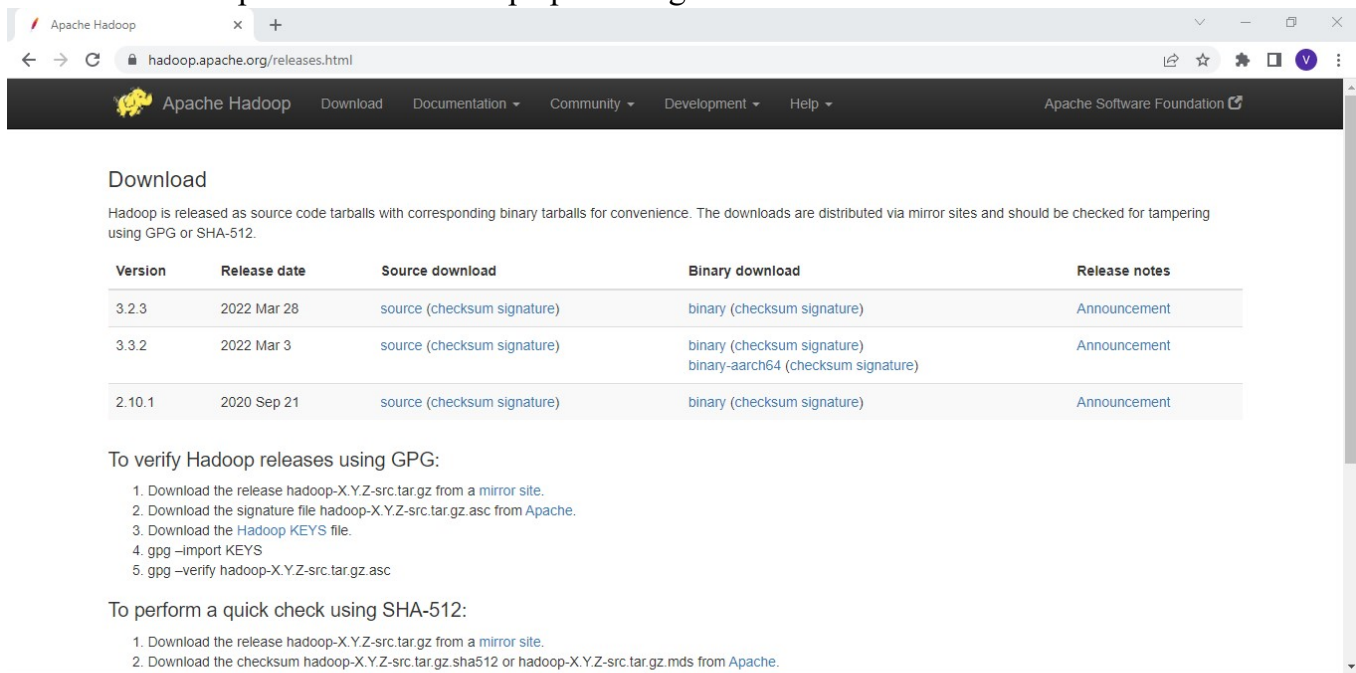
OpenJDK Runtime Environment (build 1.8.0_91-8u91-b14-3ubuntu1~16.04.1-b14)

OpenJDK 64-Bit Server VM (build 25.91-b14, mixed mode) This output verifies that OpenJDK has been successfully installed. **Note:** To set the path for environment variables. i.e. JAVA_HOME

Step2: Installing Hadoop

With Java in place, we'll visit the Apache Hadoop Releases page to find the most recent stable release. Follow the binary for the current release:

Download Hadoop from www.hadoop.apache.org



Apache Hadoop

hadoop.apache.org/releases.html

Download

Hadoop is released as source code tarballs with corresponding binary tarballs for convenience. The downloads are distributed via mirror sites and should be checked for tampering using GPG or SHA-512.

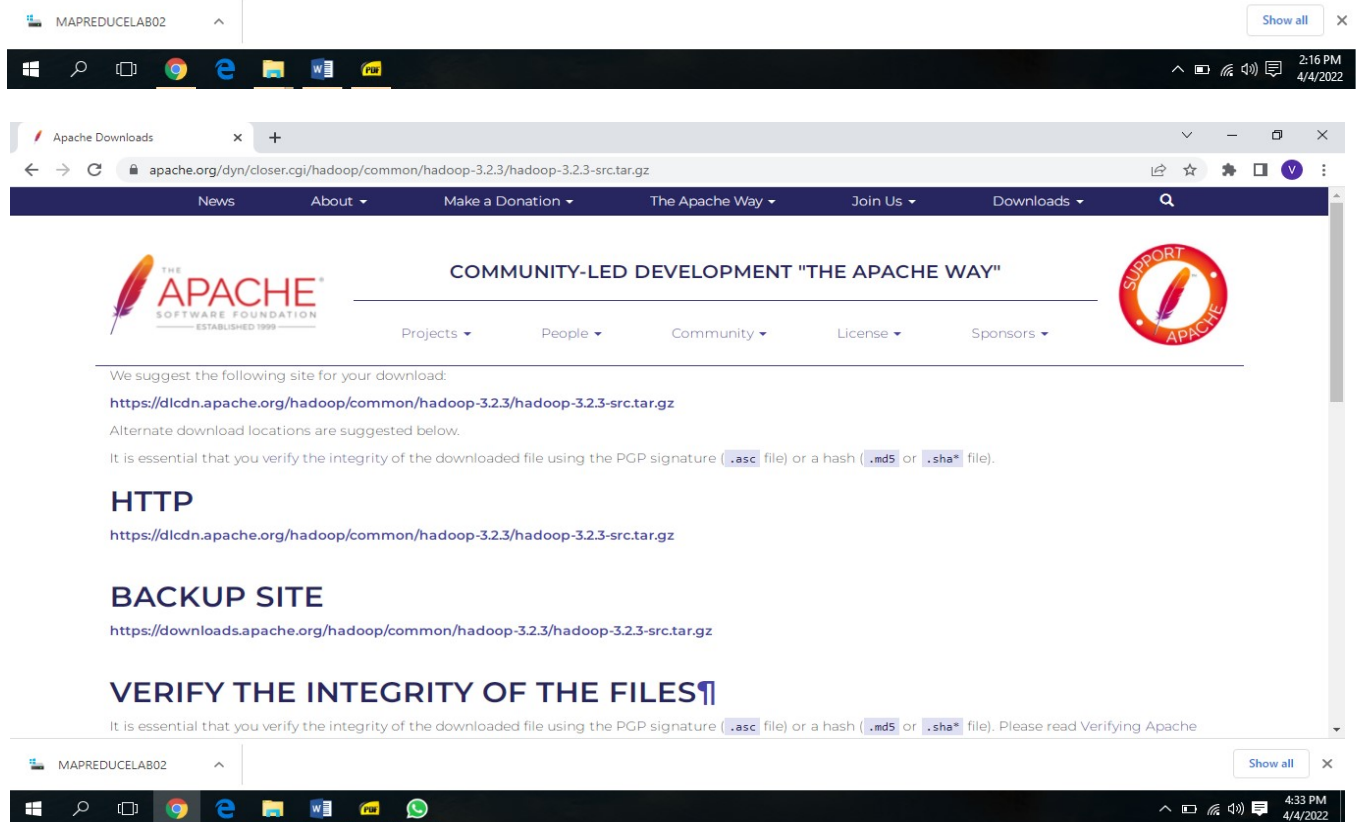
Version	Release date	Source download	Binary download	Release notes
3.2.3	2022 Mar 28	source (checksum signature)	binary (checksum signature)	Announcement
3.3.2	2022 Mar 3	source (checksum signature)	binary (checksum signature) binary-aarch64 (checksum signature)	Announcement
2.10.1	2020 Sep 21	source (checksum signature)	binary (checksum signature)	Announcement

To verify Hadoop releases using GPG:

1. Download the release `hadoop-X.Y.Z-src.tar.gz` from a [mirror site](#).
2. Download the signature file `hadoop-X.Y.Z-src.tar.gz.asc` from Apache.
3. Download the **Hadoop KEYS** file.
4. `gpg --import KEYS`
5. `gpg --verify hadoop-X.Y.Z-src.tar.gz.asc`

To perform a quick check using SHA-512:

1. Download the release `hadoop-X.Y.Z-src.tar.gz` from a [mirror site](#).
2. Download the checksum `hadoop-X.Y.Z-src.tar.gz.sha512` or `hadoop-X.Y.Z-src.tar.gz.mds` from Apache.



Apache Downloads

apache.org/dyn/closer.cgi/hadoop/common/hadoop-3.2.3/hadoop-3.2.3-src.tar.gz

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We suggest the following site for your download:

<https://d1cdn.apache.org/hadoop/common/hadoop-3.2.3/hadoop-3.2.3-src.tar.gz>

Alternate download locations are suggested below.

It is essential that you verify the integrity of the downloaded file using the PGP signature (`.asc` file) or a hash (`.md5` or `.sha*` file).

HTTP

<https://d1cdn.apache.org/hadoop/common/hadoop-3.2.3/hadoop-3.2.3-src.tar.gz>

BACKUP SITE

<https://downloads.apache.org/hadoop/common/hadoop-3.2.3/hadoop-3.2.3-src.tar.gz>

VERIFY THE INTEGRITY OF THE FILES

It is essential that you verify the integrity of the downloaded file using the PGP signature (`.asc` file) or a hash (`.md5` or `.sha*` file). Please read [Verifying Apache](#)

Procedure to Run Hadoop

1. Install Apache Hadoop 2.2.0 in Microsoft Windows OS

If Apache Hadoop 2.2.0 is not already installed then follow the post Build, Install, Configure and Run Apache Hadoop 2.2.0 in Microsoft Windows OS.

2. Start HDFS (Namenode and Datanode) and YARN (Resource Manager and Node Manager)

Run following commands.

Command Prompt

C:\Users\abhijitg>cd c:\hadoop

c:\hadoop>sbin\start-dfs

c:\hadoop>sbin\start-yarn starting yarn

daemons

Namenode, Datanode, Resource Manager and Node Manager will be started in few minutes and ready to execute Hadoop **MapReduce** job in the Single Node (pseudo-distributed mode) cluster.

```

Apache Hadoop Distribution - hadoop namenode
infoPort=50075, ipcPort=50020, storageInfo=lv=-47;cid=CID-1a1d9f-efee-4d4e-9f03-2c23e5eb;nsid=28920020;c=0) storage DS-2081780230-50010-1383498502153
13/11/03 22:38:22 INFO net.NetworkTopology: Adding a new node: /default-rack/127.0.0.1:50010
13/11/03 22:38:22 INFO blockmanagement.BlockManager: BLOCK* processReport: Received first block report from 127.0.0.1:50010 after starting up or becoming active. Its block contents are no longer considered stale
13/11/03 22:38:22 INFO BlockStateChange: BLOCK* processReport: from DatanodeRegistration(127.0.0.1, storageID=DS-2081780230-50010-1383498502153, infoPort=50075, ipcPort=50020, storageInfo=lv=-47;cid=CID-1a1d9f-efee-4d4e-9f03-2c23e5eb;nsid=28920020;c=0), blocks: 0, processing time: 15 msec

Apache Hadoop Distribution - hadoop datanode
207-50010-1383498502153> service to localhost/127.0.0.1:9000
13/11/03 22:38:22 INFO datanode.DataNode: BlockReport of 0 blocks took 0 msec to generate and 94 msec for RPC and NN processing
13/11/03 22:38:22 INFO datanode.DataNode: sent block report, processed command: org.apache.hadoop.hdfs.server.protocol.FinalizeCommand@78a6195e
13/11/03 22:38:22 INFO util.GSet: Computing capacity for map BlockMap
13/11/03 22:38:22 INFO util.GSet: VM type = 64-bit
13/11/03 22:38:22 INFO util.GSet: 0.5% max memory = 888.9 MB
13/11/03 22:38:22 INFO util.GSet: capacity = 2^19 = 524288 entries
13/11/03 22:38:22 INFO datanode.BlockPool$SliceScanner: Periodic Block Verification
  
```

Resource Manager & Node Manager:

```

C:\Apache Hadoop Distribution - yarn resourcemanager
oop.yarn.server.api.ResourceManagerAdministrationProtocolPB to the server
13/11/03 22:48:14 INFO ipc.Server: IPC Server Responder: starting
13/11/03 22:48:14 INFO ipc.Server: IPC Server listener on 8033: starting
13/11/03 22:48:14 INFO util.RackResolver: Resolved ABHIJITG. .com to /default-
rack
13/11/03 22:48:14 INFO resourcemanager.ResourceTrackerService: NodeManager from
node ABHIJITG. .com(cmPort: 60092 httpPort: 8042) registered with capability:
<memory:8192, vCores:8>, assigned nodeId ABHIJITG. .com:60092
13/11/03 22:48:14 INFO rmnode.RMNodeImpl: ABHIJITG. .com:60092 Node Transition
ed from NEW to RUNNING
13/11/03 22:48:14 INFO capacity.CapacityScheduler: Added node ABHIJITG. .com:6
0092 clusterResource: <memory:8192, vCores:8>

C:\Apache Hadoop Distribution - yarn nodemanager
13/11/03 22:48:13 INFO morthay.log: Started SelectChannelConnector00.0.0.0:8042
13/11/03 22:48:13 INFO webapp.WebApps: Web app /node started at 8042
13/11/03 22:48:14 INFO webapp.WebApps: Registered webapp guice modules
13/11/03 22:48:14 INFO client.RMPProxy: Connecting to ResourceManager at /0.0.0.0
:8031
13/11/03 22:48:14 INFO security.NMContainerTokenSecretManager: Rolling master-ke
y for container-tokens, got key with id 441918079
13/11/03 22:48:14 INFO security.NMTokenSecretManagerInNM: Rolling master-key for
nm-tokens, got key with id :1221761938
13/11/03 22:48:14 INFO nodemanager.NodeStatusUpdaterImpl: Registered with Resour
ceManager as ABHIJITG. .com:60092 with total resource of <memory:8192, vCores:
8>
13/11/03 22:48:14 INFO nodemanager.NodeStatusUpdaterImpl: Notifying ContainerMan
ager to unblock new container-requests
  
```

Run wordcount MapReduce job

Now we'll run **wordcount** MapReduce job available

in %HADOOP_HOME%\share\hadoop\mapreduce\hadoop-mapreduce-examples-2.2.0.jar

Create a text file with some content. We'll pass this file as input to the **wordcount** MapReduce job for counting words.
C:\file1.txt

Install Hadoop

Run Hadoop Wordcount Mapreduce Example

Create a directory (say 'input') in HDFS to keep all the text files (say 'file1.txt') to be used for counting words.

C:\Users\abhijitg>cd c:\hadoop C:\hadoop>bin\hdfs dfs -mkdir input

Copy the text file(say 'file1.txt') from local disk to the newly created 'input' directory in HDFS.

C:\hadoop>bin\hdfs dfs -copyFromLocal c:/file1.txt input

Check content of the copied file.

C:\hadoop>hdfs dfs -ls input

Found 1 items

-rw-r--r-- 1 ABHIJITG supergroup 55 2014-02-03 13:19 input/file1.txt

```
C:\hadoop>bin\hdfs dfs -cat input/file1.txt
```

Install Hadoop

Run Hadoop Wordcount Mapreduce Example

Run the wordcount MapReduce job provided

```
in %HADOOP_HOME%\share\hadoop\mapreduce\hadoop-mapreduce-examples-2.2.0.jar
```

```
C:\hadoop>bin\yarn jar share/hadoop/mapreduce/hadoop-mapreduce-examples-2.2.0.jar wordcount input output
```

```
14/02/03 13:22:02 INFO client.RMProxy: Connecting to ResourceManager at /0.0.0.0:8032
```

```
14/02/03 13:22:03 INFO input.FileInputFormat: Total input paths to process : 1 14/02/03
```

```
13:22:03 INFO mapreduce.JobSubmitter: number of splits:1
```

```
::
```

```
14/02/03 13:22:04 INFO mapreduce.JobSubmitter: Submitting tokens for job:
```

```
job_1391412385921_0002
```

```
14/02/03 13:22:04 INFO impl.YarnClientImpl: Submitted application
```

```
application_1391412385921_0002 to ResourceManager at /0.0.0.0:8032 14/02/03 13:22:04
```

```
INFO mapreduce.Job: The url to track the job:
```

```
http://ABHIJITG:8088/proxy/application_1391412385921_0002/
```

```
14/02/03 13:22:04 INFO mapreduce.Job: Running job: job_1391412385921_0002 14/02/03
```

```
13:22:14 INFO mapreduce.Job: Job job_1391412385921_0002 running in uber mode : false
```

```
14/02/03 13:22:14 INFO mapreduce.Job: map 0% reduce 0%
```

```
14/02/03 13:22:22 INFO mapreduce.Job: map 100% reduce 0%
```

```
14/02/03 13:22:30 INFO mapreduce.Job: map 100% reduce 100%
```

```
14/02/03 13:22:30 INFO mapreduce.Job: Job job_1391412385921_0002 completed successfully
```

```
14/02/03 13:22:31 INFO mapreduce.Job: Counters: 43
```

File System Counters

FILE: Number of bytes read=89

FILE: Number of bytes written=160142

FILE: Number of read operations=0

FILE: Number of large read operations=0

FILE: Number of write operations=0

HDFS: Number of bytes read=171

HDFS: Number of bytes written=59

HDFS: Number of read operations=6

HDFS: Number of large read operations=0

HDFS: Number of write operations=2

Job Counters

Launched map tasks=1

Launched reduce tasks=1

Data-local map tasks=1

Total time spent by all maps in occupied slots (ms)=5657

Total time spent by all reduces in occupied slots (ms)=6128

Map-Reduce Framework

Map input records=2

Map output records=7

Map output bytes=82

Map output materialized bytes=89

Input split bytes=116

Combine input records=7

Combine output records=6

Reduce input groups=6

Reduce shuffle bytes=89

Reduce input records=6

Reduce output records=6

Spilled Records=12

Shuffled Maps =1

Failed Shuffles=0

Merged Map outputs=1

GC time elapsed (ms)=145

CPU time spent (ms)=1418

Physical memory (bytes) snapshot=368246784

Virtual memory (bytes) snapshot=513716224

Total committed heap usage (bytes)=307757056

Shuffle Errors

BAD_ID=0

CONNECTION=0

IO_ERROR=0

WRONG_LENGTH=0

WRONG_MAP=0

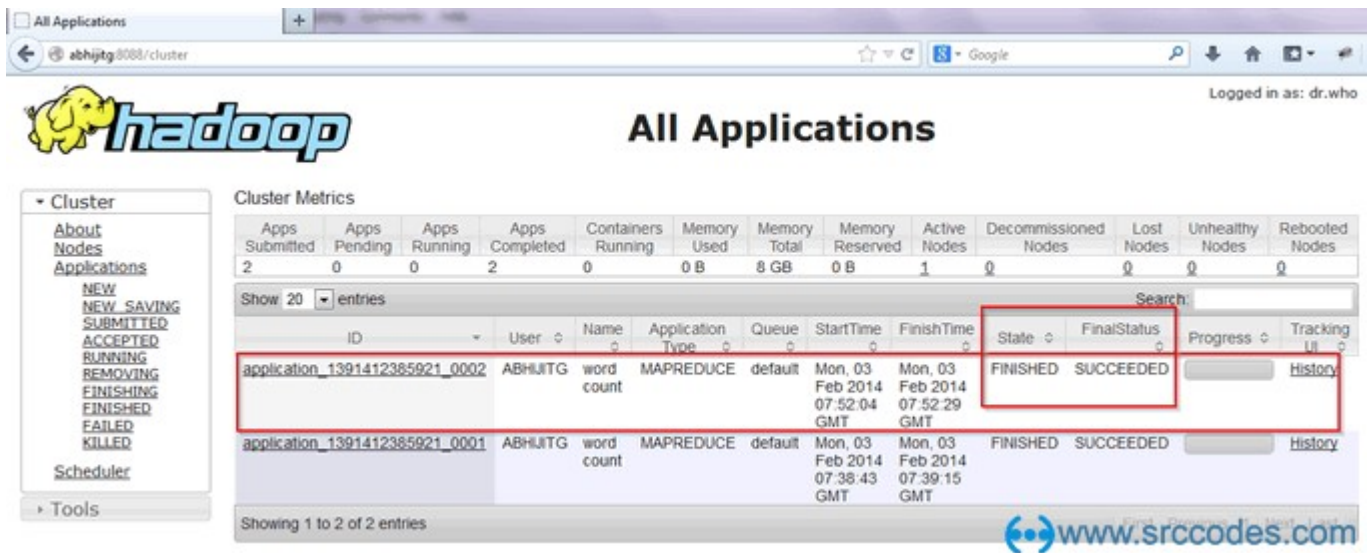
WRONG_REDUCE=0

File Input Format Counters

Bytes Read=55

File Output Format Counters

Bytes Written=59



The screenshot shows the Hadoop web interface for 'All Applications'. The browser address bar shows 'abhi1tg8088/cluster'. The page title is 'All Applications'. The Hadoop logo is on the left. A sidebar on the left contains a 'Cluster' menu with options like 'About', 'Nodes', 'Applications', 'NEW', 'NEW SAVING', 'SUBMITTED', 'ACCEPTED', 'RUNNING', 'REMOVING', 'FINISHING', 'FINISHED', 'FAILED', 'KILLED', and 'Scheduler'. The main content area shows 'Cluster Metrics' and a table of applications. The table has columns: ID, User, Name, Application Type, Queue, StartTime, FinishTime, State, FinalStatus, Progress, and Tracking UI. Two applications are listed, both with 'FINISHED' state and 'SUCCEEDED' final status. A red box highlights the 'State' and 'FinalStatus' columns for the first application. The bottom right corner has a watermark 'www.srccodes.com'.

Cluster Metrics

Apps Submitted	Apps Pending	Apps Running	Apps Completed	Containers Running	Memory Used	Memory Total	Memory Reserved	Active Nodes	Decommissioned Nodes	Lost Nodes	Unhealthy Nodes	Rebooted Nodes
2	0	0	2	0	0 B	8 GB	0 B	1	0	0	0	0

Show 20 entries

ID	User	Name	Application Type	Queue	StartTime	FinishTime	State	FinalStatus	Progress	Tracking UI
application_1391412385921_0002	ABHIJITG	word count	MAPREDUCE	default	Mon, 03 Feb 2014 07:52:04 GMT	Mon, 03 Feb 2014 07:52:29 GMT	FINISHED	SUCCEEDED		History
application_1391412385921_0001	ABHIJITG	word count	MAPREDUCE	default	Mon, 03 Feb 2014 07:38:43 GMT	Mon, 03 Feb 2014 07:39:15 GMT	FINISHED	SUCCEEDED		History

Showing 1 to 2 of 2 entries

www.srccodes.com

Result: We've installed Hadoop in stand-alone mode and verified it by running an example program it provided.